Research Task 08: LLM Bias Detection in Data Narratives

Executive Summary

This project investigates potential biases in large language model (LLM) data narratives by systematically varying prompts using identical team-level football statistics. Focusing on framing effects, proxy/structural bias, and confirmation bias, the experiment employs sanitized sports data and manually collects LLM outputs (Claude 4.5 and GPT-5) to compare how response characteristics shift under different prompt conditions. This report documents the design, data collection process, and preparation for subsequent analysis.

Introduction

Bias in artificial intelligence outputs, especially in language model–generated data summaries, can alter interpretations and recommendations even when input statistics remain constant. Our research aims to reveal if and how such biases emerge by altering prompt phrasing and context, using a consistent dataset from a prior sports analytics task. This phase covers work completed as of November 15, 2025.

Methodology

Dataset and Sanitization

- We used team-level statistics from a Premier League dataset (prior task), sanitized to replace all real club names with anonymous labels (e.g., "Team Alpha," "Team Phi" for Brentford).
- The raw dataset is not included in the repository (per instructions); regeneration instructions are included in the experiment_design notebook.

Experimental Design

- **Three hypotheses:**
    - **H1 (Framing):** Negative versus positive framing of the same team's performance.
    - **H2 (Proxy/Structural):** Neutral versus explicit "low market status" framing.
    - **H3 (Confirmation):** Neutral question versus primed with a specific conclusion (e.g., defensive strength).
- **Prompt set:** Six prompts (3 hypotheses x 2 conditions), each embedding an identical sanitized match snippet within the pair.
- Prompts and design logic are documented in prompts/all_prompts.csv (and/or .txt) and experiment_design.ipynb.

Data Collection

- Automated API access (Claude via Syracuse University credentials) was not possible; as a result, all model outputs were collected manually.
- For each condition, two samples were collected: Sample 1 from Claude 4.5 and Sample 2 from GPT-5.
- For each run, responses are recorded in a structured workbook ('results/results.xlsx') with metadata: hypothesis, condition, prompt summary, sample number, and narrative text.

Results (to-date)

- Phase 1 (design) and Phase 2 (manual collection and logging) are complete.
- The repository includes:
    - prompts/all_prompts.* — the exact prompts as run
    - notebooks/experiment_design.ipynb — prompt generation, anonymization logic
    - results/results.xlsx — LLM response samples for each prompt/condition
- No automated/quantitative analysis, claim validation, or bias visualization has yet been performed.

Limitations

- Only two LLM outputs per prompt condition are included due to the manual nature of data collection and constraints on model/API access.
- No analysis of sentiment, themes, or factual consistency is presented in this report.
- Raw and sanitized datasets are excluded from the repository; reproducibility depends on the provided code and instructions.

Next Steps

- Complete analysis of collected results (sentiment scoring, qualitative review, claim validation) as described in the repository's analysis plan.
- Compile a comprehensive bias catalogue, visual summary, and recommendations in the final report.